

## Part B Chapter 14b

### Integrating Models

#### 1 Introduction

In the previous chapter, a domain model was used to create an intelligent agent capable of analyzing a particular (human) process, providing support when necessary and adapt its behaviour when performance is non-optimal. Although converting a domain model to a model used by this intelligent agent can be accomplished by following a number of simple steps, it is important to understand the rationale behind this process (Bosse et al., 2008).

Before the reasoning techniques are covered in detail, it is important to note that any assumption made when developing the domain model is still in place when transforming the domain model into any other type of model. When developing an analysis, support or adaptation model, it is important to keep these assumptions in mind. From looking at the domain model alone it could be that certain reasoning steps seem acceptable. However, considering the assumptions made when modelling the domain another approach for analysis or support could provide more reliable results. Examples hereof will be covered in the following sections where both forward and backward reasoning will be explained in more detail.

#### 2 Forward Reasoning

Forward reasoning is a form of inductive reasoning where a conclusion is accepted based on one or more premises being true. Considering the graphical models used in this course, forward reasoning implies a reasoning technique following the directions of the arrows. In the following paragraphs, forward reasoning is applied to a number of different models.

##### ONE TO ONE

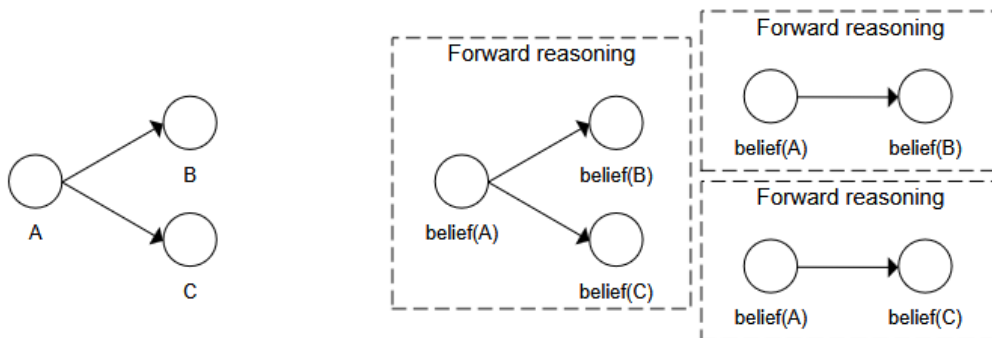
In the most simple situation, one concept leads to one other concept. Using forward reasoning, we would conclude that when the agent believes A to be true, the agent subsequently will also believe B to be true. Figure 1 (and subsequent figures below) illustrate such steps, showing in the left a relation in a domain model, while on the right the corresponding beliefs are shown as they could appear in an analysis or support model. Note that the (shorter) formalisation ‘belief(A)’ is used, which is less precise than the more commonly used ‘belief(agent, A)’.



**Figure 1.** One-to-one model, forward reasoning

### ONE TO MANY

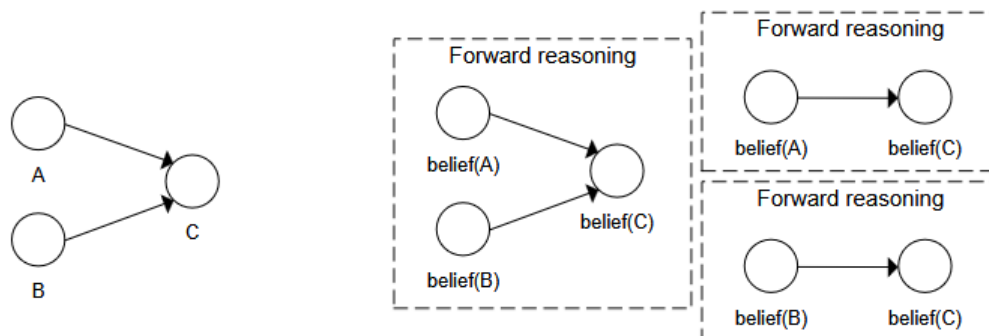
Figure 2 shows on the left a model where one concept leads to two distinct concepts. In principle, this situation is no different than the previous; A leads to B and simultaneously A also leads to C. Therefore, depending on the requirements of the model developed, various reasoning steps can be made. In its most complete form, an ambient agent believing A to be true will infer the beliefs that both B and C are also true (middle of Figure 2). However, when modelling there is the option to leave out either B or C if that knowledge is not required for the application at hand (right part of Figure 2).



**Figure 2.** One-to-many model, forward reasoning.

### MANY TO ONE

Alternatively, there could be multiple concepts leading to the same concept as shown in Figure 3 on the left. Again, this can be translated back to the simple one-to-one situation that both A leads to C and alternatively also B leads to C (middle part of Figure 3). Using forward reasoning it is possible to create a complete model using all the concepts. But even so, when not required either A or B can be ignored to create a more simple model (right part of Figure 3).



**Figure 3.** Many-to-one model, forward reasoning.

However, it could be that the model has a slightly different meaning! Instead of A leading to C and B leading to C, there is also the option that both A and B are required for C to become true. In propositional logic this would be the difference between  $A \vee B \rightarrow C$  (as shown above) and  $A \wedge B \rightarrow C$  as is shown in Figure 4, using an arc between the arrows to visualise the conjunction. This in turn also affects the reasoning of the ambient agent. For the agent to infer C as being true, it should both believe A and B to be true (middle part of Figure 4). Therefore it is not possible anymore to simply ignore A or B in this transformation of the domain model. However, if it is not possible to generate a belief about both concepts, for example due to the lack of an appropriate sensor for one of them, setting a default value could be considered. Keep in mind however, that with such assumptions, the model will become less and less reliable as it in essence guesses the state of those concepts. Nevertheless, in some situations it could be argued that such a choice is acceptable, which might be possible in this example as there is still information available about either B or A (right part of Figure 4).

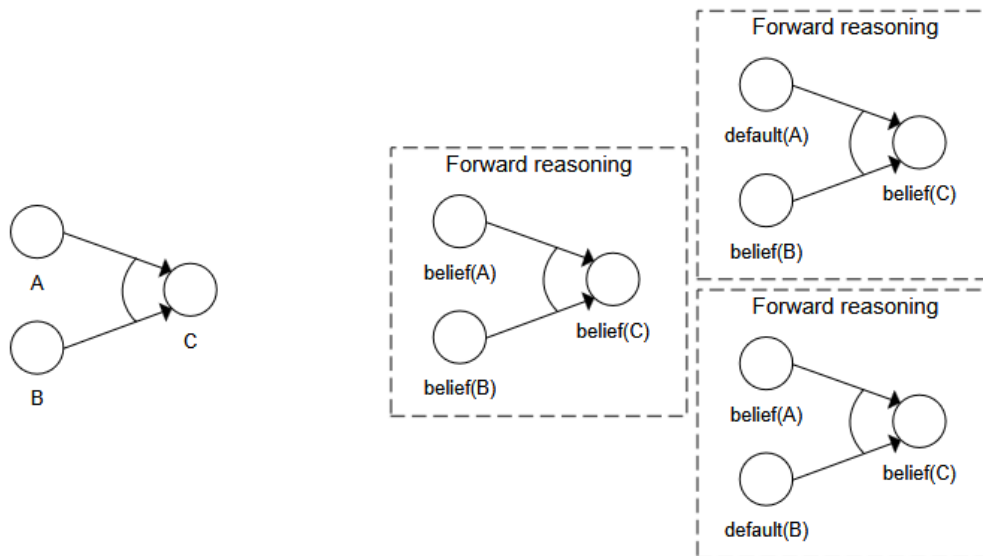


Figure 4. Many-to-one model using conjunction, forward reasoning.

### 3 Backward reasoning

Backward reasoning is a form of abduction which in essence is an inference from an effect to its cause (Josephson and Josephson, 1996). In the graphical models used here, this is accomplished by reasoning against the direction of the arrows. Care should be taken when using backward reasoning, as often there are multiple reasons for some concept to be true besides those that are modelled.

Sometimes, that what has been modelled may not even be the most plausible cause for some particular effect. That being said, backward reasoning could still be the method of choice as

often the effects of a process are more easily observed than the causes. The following sections will illustrate backward reasoning in different situations and illustrate the pitfalls that need to be considered when applying backward reasoning.

### ONE TO ONE

Figure 5 shows the simple case where A leads to B. Using backward reasoning, this would result in believing A whenever we believe B. To illustrate the risks involved with using backward reasoning, consider an equivalent statement using propositional logic:  $A \rightarrow B$ . Concluding A on the basis of B would be a logical fallacy of affirming the consequent. There are many examples that illustrate this fallacy. If it rains, I get wet; but when I'm wet it does not necessarily rain. If I'm British, I speak English; however when I speak English I'm not necessarily British. It would be logically correct to conclude not A on the basis of not having B. If I don't get wet, it doesn't rain and if I don't speak English, I'm not British.<sup>1</sup> Using backward reasoning however, we do not use this logically correct implication. Here, when the agent believes B to be true, A is considered the most probable explanation and is therefore also believed to be true. As mentioned before, it is very important to consider whether such an assumption is reasonable in the context where the ambient agent will be used!



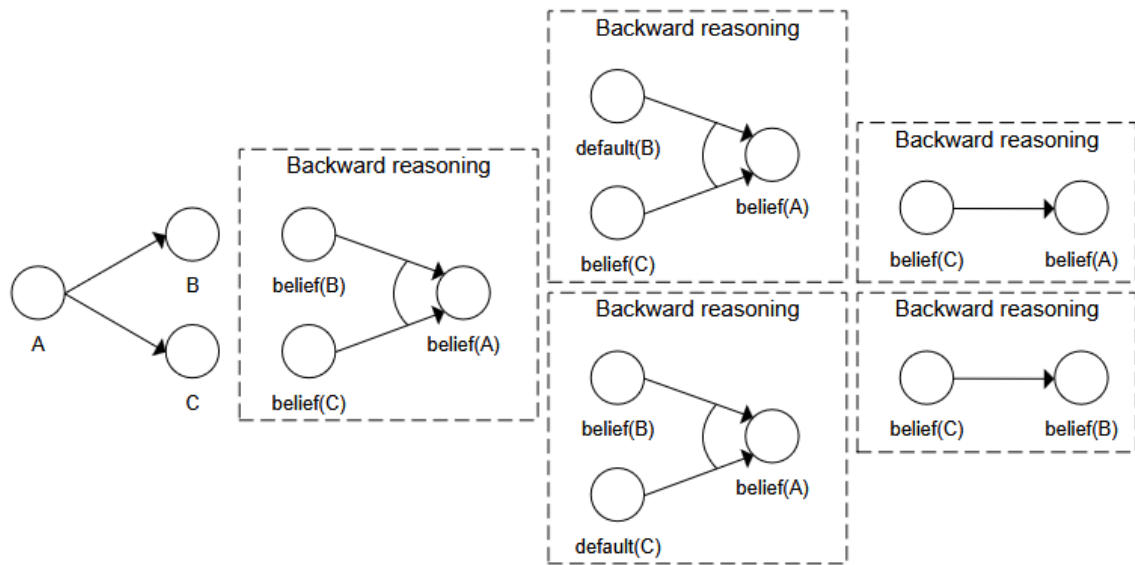
Figure 5. One-to-one model, backward reasoning.

### ONE TO MANY

If one concept leads to several other distinct concepts, as shown in Figure 6, there is a little catch with backward reasoning. Consider A being true; in that case both B and C become true. Therefore, to believe A using backward reasoning, the combination of B and C being true is a good indication A was probably the cause. Thus when B and C are believed to be true, it is reasonable to believe A to be true (leftmost reasoning model of Figure 6). Even so, if it is possible to set a default value for either B or C, it is possible to believe A only on the basis of either believing C or B respectively (middle reasoning models of Figure 6). In very few cases, it might even be possible to completely ignore B or C. If it can be argued that only believing B or C to be true is sufficient information to believe A was the cause, this can be a valid backward reasoning step (rightmost reasoning model of Figure 6).<sup>2</sup>

<sup>1</sup> There are many exceptions for these simple examples where this does not hold, for example when I use an umbrella or when I am a British baby. However, for illustrative purposes these examples should suffice.

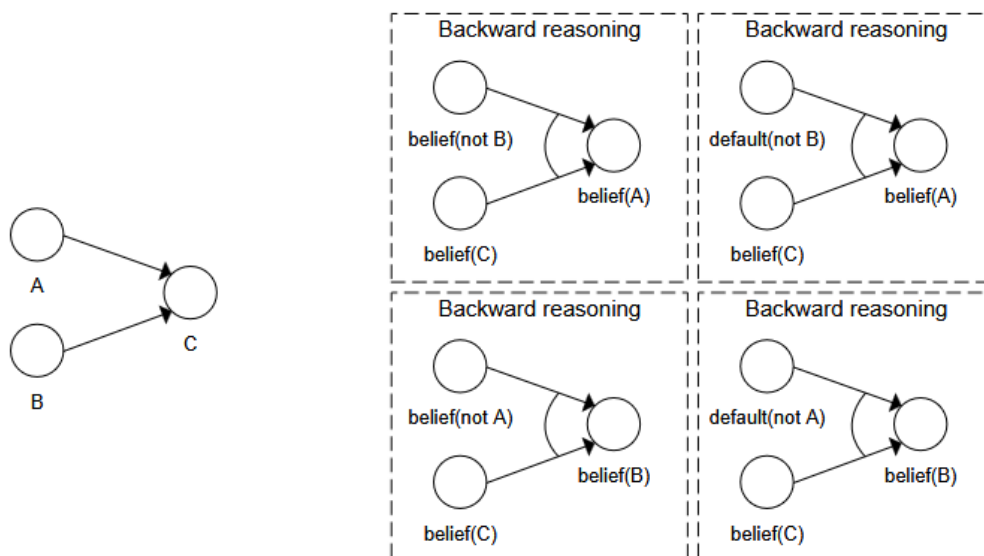
<sup>2</sup> Combining this backward reasoning step with forward reasoning can also provide a belief about C or B respectively.



**Figure 6.** One-to-many model, backward reasoning.

### MANY TO ONE

Backward reasoning becomes more complicated when multiple concepts lead to the same concept. Figure 7 shows this case where A and B independently of each other cause C. Thus, when we believe C to be true, it is not known whether this is caused by A or B. When we want to conclude A on the basis of believing C, it is also necessary to believe that B is not true. Similarly, when we believe C to be true and A to be not true, we may believe B to be true (middle part of Figure 7). Even so, if possible, by default value it can be assumed that either A or B is not true, making it possible to reason with only a belief about C. In contrast with the previous situation, we cannot ignore either A or B. With only a belief about C, there will always be uncertainty whether A or B was the cause. Thus more information is required before an agent is able to make any reasoning steps (right part of Figure 7).



**Figure 7.** Many-to-one model, backward reasoning.

When A and B are both required for C to become true, as shown in Figure 3.8, the possibilities for backward reasoning resemble more the one-to-many case. When the ambient agent believes C to be true and also believes B to be true, it is very likely A should also be true (as the combination of A and B leads to C being true). A similar reasoning can be done for concluding B (leftmost reasoning models of Figure 8). Again, if reasonable default values can be chosen for either A or B, this makes reasoning easier (middle reasoning models of Figure 8). Since it is the combination of A and B which makes C true, having only a belief about C can be sufficient to make inferences about A and/or B (rightmost reasoning models of Figure 8). However, the same warning as in the one-to-one scenario holds here; make sure there are no other plausible causes for C to be true (that are perhaps not modelled) besides the combination of A and B. Otherwise, the agent might be prone to errors by ignoring those possible explanations for C.

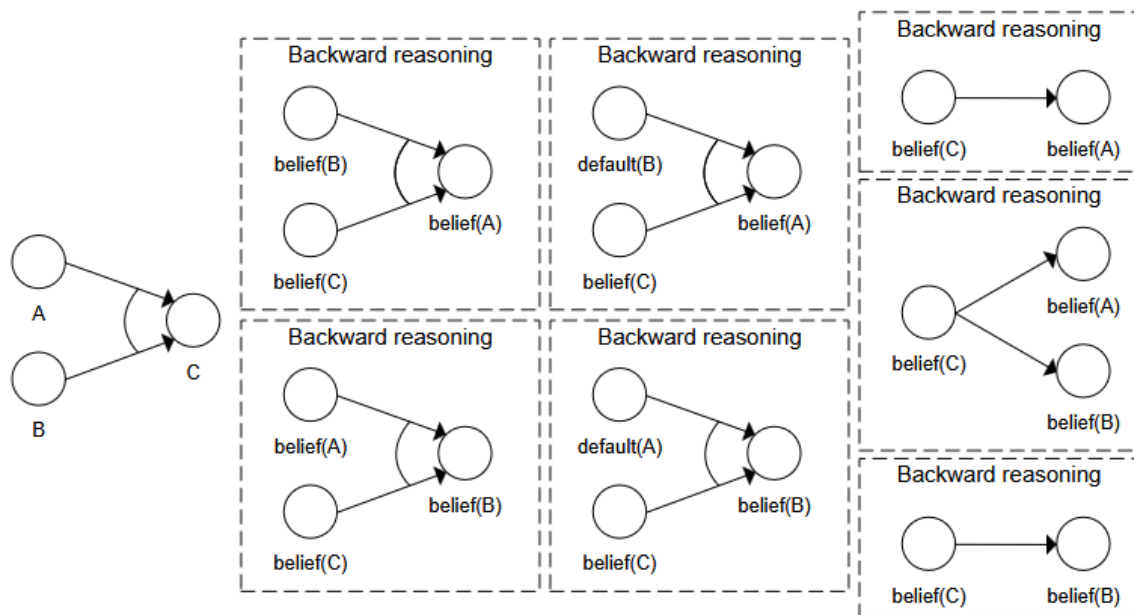


Figure 8. Many-to-one model using conjunction, backward reasoning.

#### 4 Choosing between forward reasoning and backward reasoning

An important choice that needs to be made when developing analysis and support models is that of either forward or backward reasoning. Do not make this choice lightly, because the functioning of the agent heavily depends on which technique is chosen. In case of the analysis model, the possible observations an agent can make already decide for you, but often there are options for using both techniques.

Important factors here are the considerations mentioned in Section 2.2 of Chapter 13b on input for intelligent agents. But there are more aspects to consider. An analysis model reasoning forward in time will, with a little more work, be able to make predictions about the future. Backward reasoning on the other hand can only provide an agent with an analysis later

on in the process, as the reasoning tries to find the causes for particular effects. Similarly, forward reasoning in a support model can make predictions about the effects of a particular support model, which is difficult to do using backward reasoning. In both models, backward reasoning is easier to implement; however, when using backward reasoning there is the risk of introducing logical fallacies. It is in those cases very important to carefully consider the assumptions that underly the conclusions that are drawn. Therefore, do not only consider the possible input for your agent, also keep in mind the strengths and weaknesses of both reasoning techniques.

## References

- Bosse, T., Both, F., Gerritsen, C., Hoogendoorn, M., & Treur, J. (2008). Model-based reasoning methods within an ambient intelligent agent model. In *Constructing Ambient Intelligence* (pp. 352-370). Springer Berlin Heidelberg.
- Josephson, J.R. and Josephson, S.G. (eds.) (1996), *Abductive Inference: Computation, Philosophy, Technology*. New York: Cambridge University Press.
- Wooldridge, M., & Jennings, N. R. (1995). Intelligent agents: Theory and practice. *Knowledge engineering review*, 10(2), 115-152.